MATH 202 Your name: _____

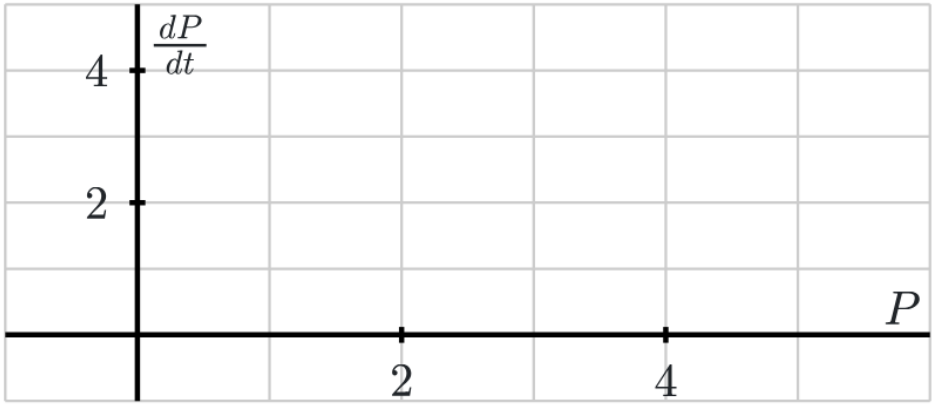
Learning target D2, version 1

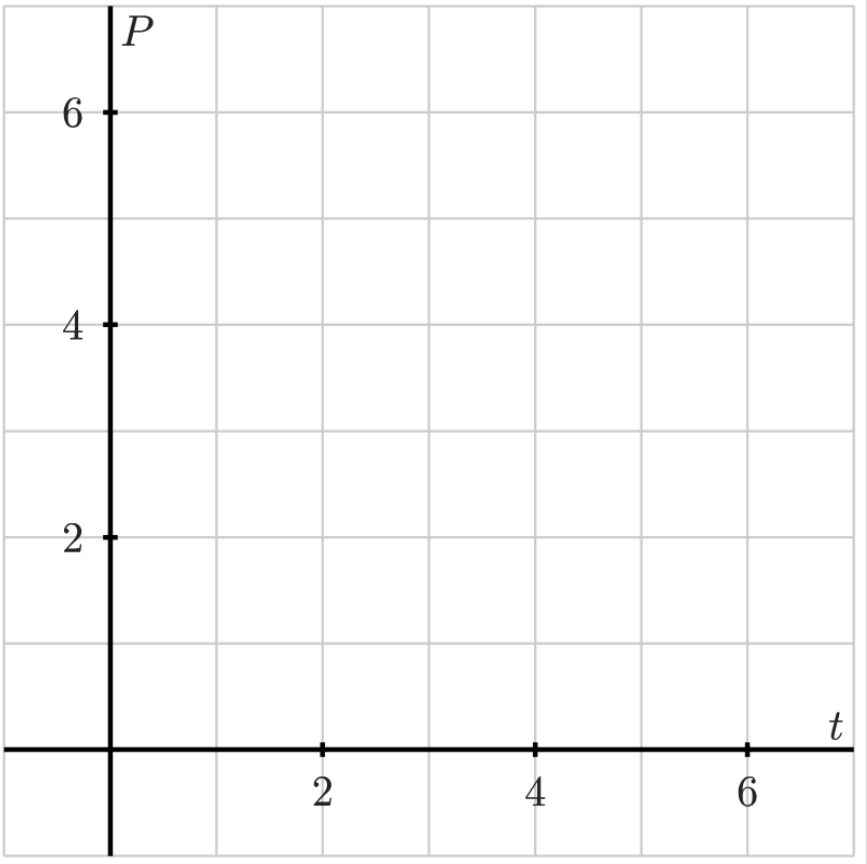
D2: Express real world situations as differential equations, determine any equilibrium solutions, and explain the significance of the equilibrium solutions in context.

A population of flamingos changes over time according to the differential equation

$$\frac{dP}{dt} = -\frac{1}{3}P(1-P)(5-P),$$

where P is measured in thousands (and t is probably measured in months or years).
Draw a phase graph and a slope field. Find all the equilibrium solutions, and explain how you know each one is stable or unstable.





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Learning target D2, version 2

D2: Express real world situations as differential equations, determine any equilibrium solutions, and explain the significance of the equilibrium solutions in context.

Newton's law of cooling says that a hot object cools[1] at a rate proportional to the difference between its current temperature and the ambient temperature, which let's say is 73 degrees:

$$\frac{dT}{dt} = -k \cdot (T - 73)$$

Draw a slope field (on T vs. t axes), a phase graph (on $\frac{dT}{dt}$ vs. T axes), or work in other ways to:

- figure out any equilibrium solution(s),
- classify the stability of any equilibrium solution(s),
- and explain why the equilibrium solution(s) make real-life sense.

[1] Or a cold object warms!